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ROLL NO : 23I-0058

SECTION : BSE – 6A

AI LAB 04

BSSE SPRING 2026

LAB 04

Task 1: Streamlit-Based Exploratory Data Analysis Interface

Develop an interactive GUI system for exploratory data analysis (EDA) using Streamlit. The system should facilitate dataset ingestion, metadata inspection, and primary visual pattern recognition, providing end-users with a structured interface for initial data exploration.

Implementation Requirements:

1. Dataset Ingestion & Metadata Inspection

- **File Upload Interface:** Implement a secure file upload module supporting CSV format.

Validate that the uploaded file is correctly formatted.

- **Dataset Overview Panel:** Display key metadata including:

- o Dimensional properties (number of rows and columns)
- o Column-wise data types
- o Count of missing values per attribute
- o Basic statistical summary for numerical attributes (mean, median, min, max)

Note: Test your implementation using the titanic.csv dataset.

2. Attribute Selection & Classification

- **Dynamic Feature Selector:** Provide a dropdown to select a single column for analysis.
- **Automated Attribute Typing:** Detect whether the selected column is numerical or categorical, and update the visualization module accordingly.

3. Visualization Module

- **Conditional Rendering:**
 - o **Numerical Attributes:** Generate a histogram depicting the value distribution. Include axis labels and a descriptive title.
 - o **Categorical Attributes:** Generate a bar chart representing frequency counts of unique values. Include axis labels and percentage display (optional).
- Ensure that visualizations are updated dynamically upon attribute selection.

4. Interface Layout & Organization

- **Sidebar Panel:** Should have all interactive controls, including file upload and column selection.
- **Main Content Area:**
 - o Top section: Dataset preview (first 5 rows) and metadata display

o Bottom section: Conditional visualization canvas

```
C:\Windows\System32\cmd.exe - streamlit run i230058_lab04_1.py
Traceback (most recent call last):
  File "C:\Users\Shah Faisal\AppData\Local\Programs\Python\Python313\Lib\site-packages\streamlit\dataframe_util.py", line 839, in convert_pandas_df_to_arrow_bytes
    table = pa.Table.from_pandas(df)
  File "pyarrow/table.pxi", line 4796, in pyarrow.lib.Table.from_pandas
  File "C:\Users\Shah Faisal\AppData\Local\Programs\Python\Python313\Lib\site-packages\pyarrow\pandas_compat.py", line 651, in dataframe_to_arrays
    arrays = [convert_column(c, f)
              ~~~~~^~~~~~
  File "C:\Users\Shah Faisal\AppData\Local\Programs\Python\Python313\Lib\site-packages\pyarrow\pandas_compat.py", line 639, in convert_column
    raise e
  File "C:\Users\Shah Faisal\AppData\Local\Programs\Python\Python313\Lib\site-packages\pyarrow\pandas_compat.py", line 633, in convert_column
    result = pa.array(col, type=type_, from_pandas=True, safe=safe)
  File "pyarrow/array.pxi", line 365, in pyarrow.lib.array
  File "pyarrow/array.pxi", line 91, in pyarrow.lib._ndarray_to_array
  File "pyarrow/error.pxi", line 92, in pyarrow.lib.check_status
pyarrow.lib.ArrowInvalid: ("Could not convert dtype('int64') with type numpy.dtypes.Int64Type: did not recognize Python value type when inferring an Arrow data type", 'Conversion failed for column 0 with type object')
Stopping...

C:\Users\Shah Faisal\Music\Semester 06\7.Artificial Intelligence Lab - 1\LAB 04>streamlit run i230058_lab04_1.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://172.17.48.88:8501
```

```
import seaborn as sns
import streamlit as st
import matplotlib.pyplot as plt
import pandas as pd
st.set_page_config(page_title="Data Analysis Platform", layout="expanded")
st.title("Data Exploration Dashboard")
st.write("Import a CSV dataset to commence analytical examination")
st.sidebar.header("Analysis Parameters")
uploaded_data = st.sidebar.file_uploader("Import CSV Dataset", type=["csv"])
if uploaded_data is not None:
    dataset = pd.read_csv(uploaded_data)
    st.subheader("Dataset Preview")
    st.dataframe(dataset.head())
    row_count, column_count = dataset.shape
    st.write(f"Total Observations: {row_count}")
    st.write(f"Total Variables: {column_count}")
    st.write("Variable Type Information:")
    st.dataframe(dataset.dtypes.rename('Data Type'))
    st.write("Null Value Distribution:")
    null_summary = dataset.isnull().sum()
    st.dataframe(null_summary[null_summary > 0].rename('Missing Count'))
    if null_summary.any() else "No missing values detected"
    st.write("Statistical Overview:")
    st.dataframe(dataset.describe())
```

```
st.subheader("Data Visualization")
selected_variable = st.selectbox(
    "Choose Variable for Graphical Representation",
    dataset.columns,
    key="var_selector"
)
if pd.api.types.is_numeric_dtype(dataset[selected_variable]):
    visualization_figure, axis = plt.subplots(figsize=(10, 6))
    axis.hist(dataset[selected_variable].dropna(),
        bins='auto',
        edgecolor='black',
        alpha=0.7)
    axis.set_title(f"Distribution Analysis: {selected_variable}",
        fontweight='bold',
        pad=15)
    axis.set_xlabel(selected_variable, fontweight='bold')
    axis.set_ylabel("Observation Frequency", fontweight='bold')
    axis.grid(axis='y', alpha=0.3)
    st.pyplot(visualization_figure)
else:
    value_frequencies = dataset[selected_variable].value_counts()
    visualization_figure, axis = plt.subplots(figsize=(10, 6))
    value_frequencies.plot(kind='bar',
        ax=axis,
        color='steelblue',
        edgecolor='black')
    axis.set_title(f"Categorical Distribution: {selected_variable}",
        fontweight='bold',
        pad=15)
    axis.set_xlabel(selected_variable, fontweight='bold')
    axis.set_ylabel("Category Count", fontweight='bold')
    axis.tick_params(axis='x', rotation=45)
    axis.grid(axis='y', alpha=0.3)
    st.pyplot(visualization_figure)
```

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Deploy ⋮

User Controls

Upload csv file

Drag and drop file here

Limit 200MB per file • CSV

Browse files

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Titanic-Dataset.csv

59.8KB

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Exploratory Data Analysis (EDA)

Upload a csv file to start analysis

Dataset Review

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ti
0	1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC
2	3	1	3	Heikkinen, Miss. Laina	female	26	0	0	ST
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	11
4	5	0	3	Allen, Mr. William Henry	male	35	0	0	37

Rows: 891

Columns: 12

Columns DataTypes:

localhost:8501

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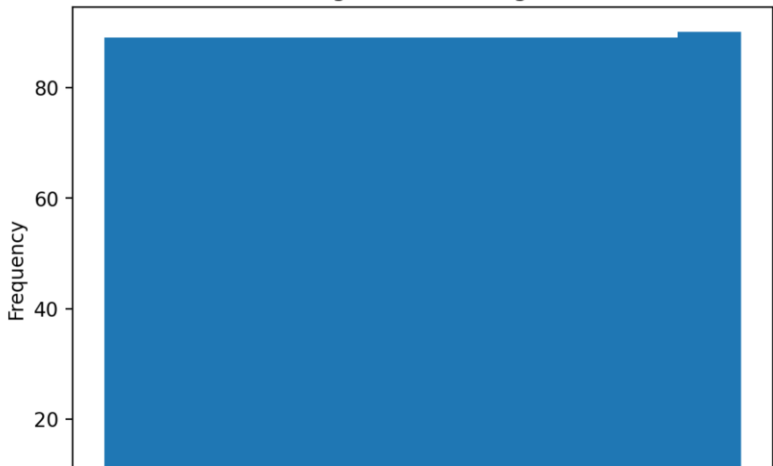
Titanic-Dataset.csv

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Visualization

Histogram of PassengerId



Deploy

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Titanic-Dataset.csv
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Visualization



Bar Chart of Sex

